

4033/5033 Assignment 3

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Due on Oct 10, 2024

Note: This assignment has four sections. Each section covers one classification technique. Some sections have both theory and programming tasks; some only have theory or programming tasks.

Part I: Logistic Regression

Theory Task The following logistic regression model defines the posterior differently from what we studied in class.

$$P(Y = 1 \mid X = x) = \frac{\exp(-x^T \beta)}{1 + \exp(-x^T \beta)}, \quad (1)$$

and

$$P(Y = 0 \mid X = x) = \frac{1}{1 + \exp(-x^T \beta)}. \quad (2)$$

Derive the matrix forms of the following derivatives:

$$\frac{\partial L(\beta)}{\partial \beta} = \quad (3)$$

$$\frac{\partial^2 L(\beta)}{\partial \beta \partial \beta^T} = \quad (4)$$

Programming Task Implement two versions of the above logistic regression model:

1. One optimized using gradient descent (pick λ by yourself), i.e. $\beta = \beta - \lambda \cdot \frac{\partial J(\beta)}{\partial \beta}$
2. The other optimized using Newton's method, i.e. $\beta = \beta - \frac{\partial^2 J(\beta)}{\partial \beta \partial \beta^T}$

Show the testing error of your model versus the number of β updates in Figure 1. The figure should contain two curves — one based on gradient descent and the other based on Newton's method. A template 'hw3Pythonlogit.py' is provided.

First Derivative: $\frac{\partial L(\beta)}{\partial \beta}$

For a single data point (x, y) , the log-likelihood is:

$$\ell(\beta) = y \log \left(\frac{e^{-x^T \beta}}{1 + e^{-x^T \beta}} \right) + (1 - y) \log \left(\frac{1}{1 + e^{-x^T \beta}} \right)$$

Simplifying:

$$\ell(\beta) = -yx^T \beta - \log(1 + e^{-x^T \beta})$$

Taking the derivative with respect to β :

$$\frac{\partial \ell}{\partial \beta} = -yx + \frac{e^{-x^T \beta}}{1 + e^{-x^T \beta}} \cdot x = \left(\frac{e^{-x^T \beta}}{1 + e^{-x^T \beta}} - y \right) x$$

For n training examples, summing over all data points:

$$\boxed{\frac{\partial L(\beta)}{\partial \beta} = \sum_{i=1}^n \left(\frac{e^{-x_i^T \beta}}{1 + e^{-x_i^T \beta}} - y_i \right) x_i = X^T (p - y)}$$

where $p_i = P(Y = 1|x_i) = \frac{e^{-x_i^T \beta}}{1 + e^{-x_i^T \beta}}$.

Second Derivative: $\frac{\partial^2 L(\beta)}{\partial \beta \partial \beta^T}$

Let $p = \frac{e^{-x^T \beta}}{1 + e^{-x^T \beta}}$. We need $\frac{\partial p}{\partial \beta}$:

$$\frac{\partial p}{\partial \beta} = -\frac{e^{-x^T \beta}}{(1 + e^{-x^T \beta})^2} \cdot x = -p(1 - p)x$$

where $1 - p = \frac{1}{1 + e^{-x^T \beta}}$.

Now, taking the second derivative of $\ell(\beta) = (p - y)x$:

$$\frac{\partial^2 \ell}{\partial \beta \partial \beta^T} = x \cdot \frac{\partial p}{\partial \beta^T} = -p(1 - p) \cdot xx^T$$

For n training examples:

$$\boxed{\frac{\partial^2 L(\beta)}{\partial \beta \partial \beta^T} = -\sum_{i=1}^n p_i(1 - p_i) \cdot x_i x_i^T = -X^T W X}$$

where $W = \text{diag}(p_1(1 - p_1), \dots, p_n(1 - p_n))$ is an $n \times n$ diagonal matrix.

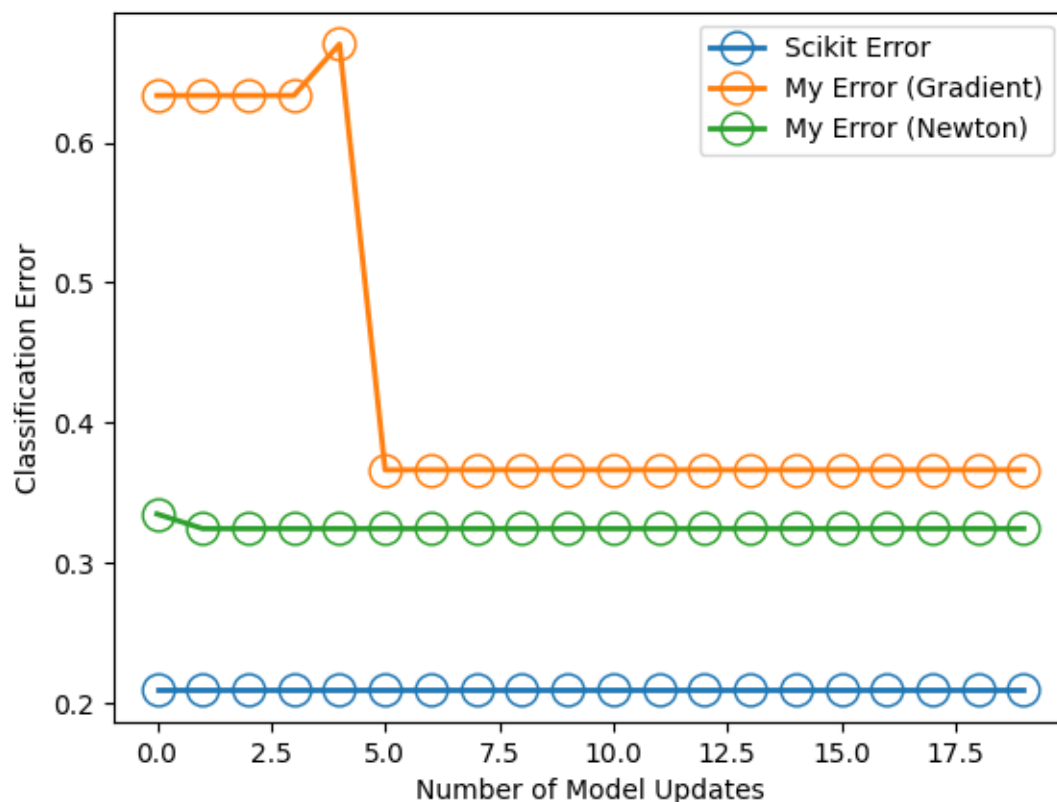


Fig. 1. Testing Error versus Model Updates

Section II: Naive Bayes

Theory Task (Only) Table 1 contains 10 students, each described by two features: ‘Course’ and ‘Hour’, and one binary label ‘GPA’. For the two features, value ‘1’ means ‘yes’ and value ‘0’ means ‘no’.

Learn a naive Bayes classifier based on Table 1 and use it to predict the GPA for a student who takes less than 3 courses and studies less than 2 hours per day. Elaborate on your arguments, including showing your estimates of all important probabilities.

We need to predict the GPA for a student with Course = 0 (takes ≤ 3 courses) and Hour = 0 (studies ≤ 2 hours per day).

Step 1: Calculate Prior Probabilities

From Table 1:

- Students with GPA = A: 4 (students 04, 06, 09, 10)
- Students with GPA = B: 6 (students 01, 02, 03, 05, 07, 08)

$$P(\text{GPA} = A) = \frac{4}{10}$$

Table 1. Student Data Set

Student ID	Take more than 3 courses?	Study more than 2 hour per day?	GPA (A/B)
01	1	1	B
02	1	0	B
03	1	0	B
04	0	0	A
05	1	1	B
06	0	1	A
07	0	1	B
08	1	1	B
09	1	1	A
10	1	0	A

$$P(\text{GPA} = \text{B}) = \frac{6}{10}$$

Step 2: Calculate Likelihoods

For GPA = A:

- Students with Course = 0: 2 out of 4 (students 04, 06)
- Students with Hour = 0: 2 out of 4 (students 04, 10)

$$P(\text{Course} = 0 | \text{GPA} = \text{A}) = \frac{2}{4}$$

$$P(\text{Hour} = 0 | \text{GPA} = \text{A}) = \frac{2}{4}$$

For GPA = B:

- Students with Course = 0: 1 out of 6 (student 07)
- Students with Hour = 0: 2 out of 6 (students 02, 03, 07)

$$P(\text{Course} = 0 | \text{GPA} = \text{B}) = \frac{1}{6}$$

$$P(\text{Hour} = 0 | \text{GPA} = \text{B}) = \frac{2}{6}$$

Step 3: Apply Naive Bayes Formula

Using the naive assumption of conditional independence:

$$\begin{aligned}
 P(\text{GPA} = \text{A} | \text{C}=0, \text{H}=0) &\propto P(\text{C}=0 | \text{A}) \times P(\text{H}=0 | \text{A}) \times P(\text{A}) \\
 &= \frac{2}{4} \times \frac{2}{4} \times \frac{4}{10} = \frac{16}{160} = \frac{1}{10}
 \end{aligned}$$

$$\begin{aligned}
 P(\text{GPA} = \text{B} | \text{C}=0, \text{H}=0) &\propto P(\text{C}=0 | \text{B}) \times P(\text{H}=0 | \text{B}) \times P(\text{B}) \\
 &= \frac{1}{6} \times \frac{2}{6} \times \frac{6}{10} = \frac{12}{360} = \frac{1}{30}
 \end{aligned}$$

Step 4: Make Prediction

Since $0.1 > 0.033$, we predict:

$$\boxed{\text{GPA} = \text{A}}$$

A student who takes less than 3 courses and studies less than 2 hours per day is predicted to have a GPA of A.

Section III: k-Nearest Neighbor

Programming Task (Only)

Task 1. Implement kNN. Plot two curves in Figure 2: - One for training error versus k - One for testing error versus k

Pick 5 values of k by yourself.

Note: You need to write code to: - Measure the distance between testing instances and training instances - Find k nearest neighbors based on the distance - Implement majority vote by the neighbors

We will no longer include scikit-learn implementation as a baseline in this and most future assignments. But feel free to use it as a reference to check correctness of your implementation. A template `hw3_python_knn.py` is provided.

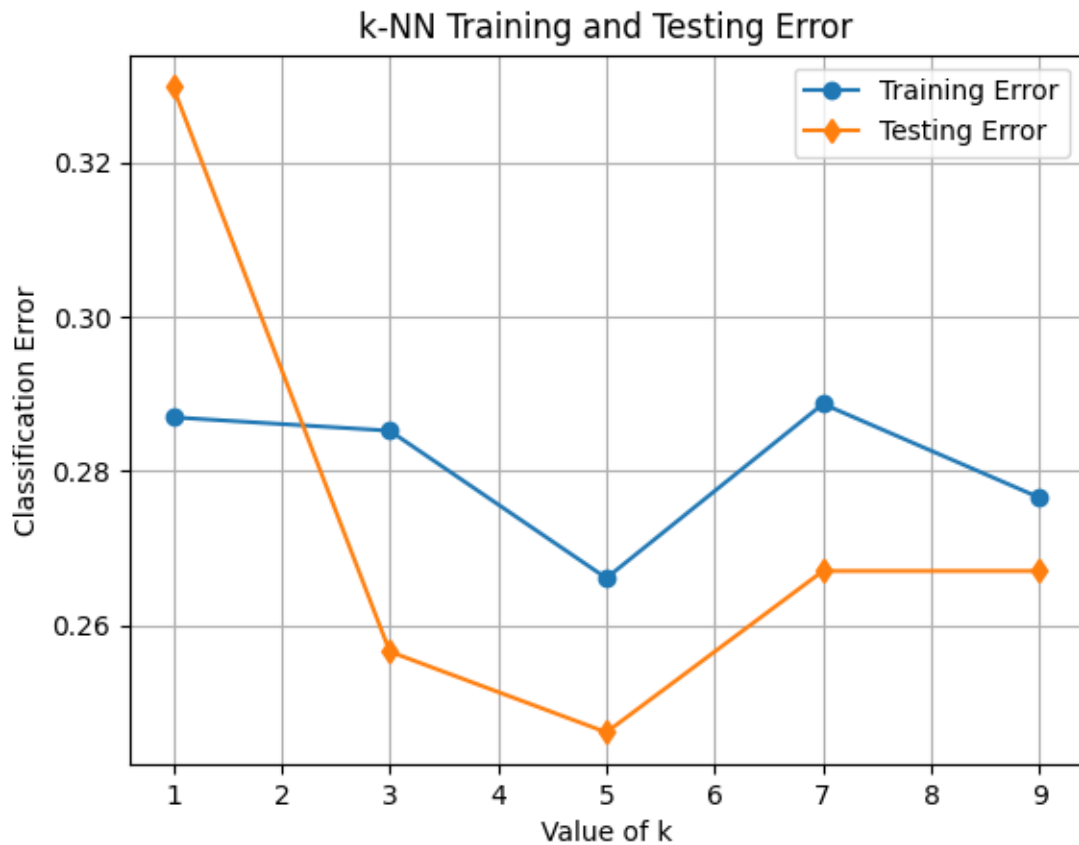


Fig. 2. Testing Error versus k

Task 2. Consider a compressed kNN (CkNN) model that first compresses the initial training set of n instances into a random subset of m instances, and then performs kNN classification based on the compressed set.

Your task is to implement CkNN. Plot a curve of its testing error versus compression rate $\frac{m}{n}$ in Figure 3. Pick a value for k and fix it. Pick 5 values of m by yourself — but the last value must be n .

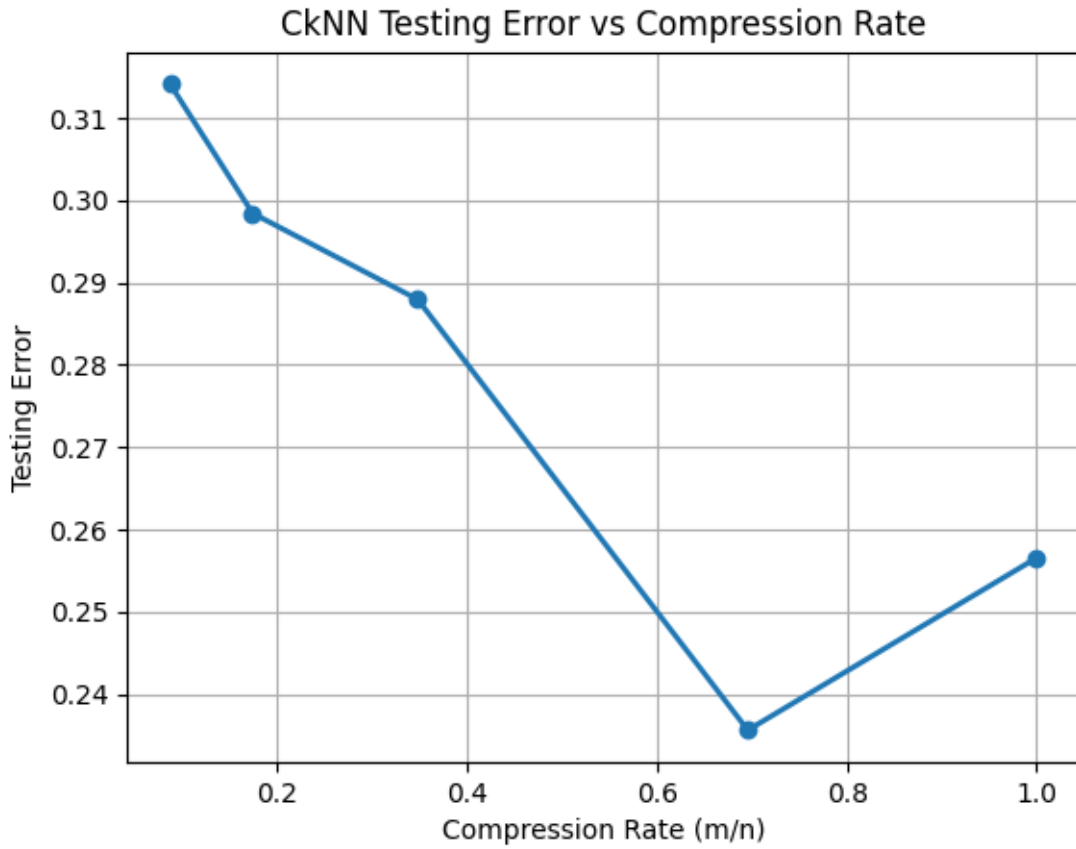


Fig. 3. Testing Error versus Compression Rate

Part IV: Decision Tree

Theory Task (Only)

Table 2 contains the records of ten students, described by five features (F1, F2, F3, F4, F5) and one binary label ‘Grade’.

Suppose we want to learn a decision tree model from this table. Your task is to identify the best feature to split the root node. You need to elaborate your arguments — especially how to estimate the label entropy of split child nodes based on each candidate feature.

Parent Entropy Calculation Before any split, we have:

- Total students: 10
- Grade A: 4 students (04, 06, 09, 10)
- Grade B: 6 students (01, 02, 03, 05, 07, 08)
- $p_A = \frac{4}{10} = 0.4$, $p_B = \frac{6}{10} = 0.6$

$$\begin{aligned}
 H_{\text{parent}} &= -(0.4) \log_2(0.4) - (0.6) \log_2(0.6) \\
 &= -(0.4)(-1.322) - (0.6)(-0.737) \\
 &= 0.529 + 0.442 = 0.971
 \end{aligned}$$

F1 Split Analysis

Table 2. Student Data Set

Student ID	F1	F2	F3	F4	F5	Grade (A/B)
01	1	1	0	0	1	B
02	1	0	0	1	1	B
03	1	0	1	0	0	B
04	0	0	1	1	0	A
05	1	1	0	1	0	B
06	0	1	0	1	1	A
07	0	1	1	1	1	B
08	1	1	0	0	1	B
09	1	1	1	0	1	A
10	1	0	1	0	0	A

Split based on F1:

F1 = 1 (students: 01, 02, 03, 05, 08, 09, 10):

- Grade A: 2 (students 09, 10)
- Grade B: 5 (students 01, 02, 03, 05, 08)
- Total: 7 students
- $p_A = \frac{2}{7} = 0.286$, $p_B = \frac{5}{7} = 0.714$

$$\begin{aligned}
 H_{F1=1} &= -(0.286) \log_2(0.286) - (0.714) \log_2(0.714) \\
 &= -(0.286)(-1.807) - (0.714)(-0.486) \\
 &= 0.517 + 0.347 = 0.863
 \end{aligned}$$

F1 = 0 (students: 04, 06, 07):

- Grade A: 2 (students 04, 06)
- Grade B: 1 (student 07)
- Total: 3 students
- $p_A = \frac{2}{3} = 0.667$, $p_B = \frac{1}{3} = 0.333$

$$\begin{aligned}
 H_{F1=0} &= -(0.667) \log_2(0.667) - (0.333) \log_2(0.333) \\
 &= -(0.667)(-0.585) - (0.333)(-1.585) \\
 &= 0.390 + 0.528 = 0.918
 \end{aligned}$$

Weighted Average Entropy After F1 Split

$$\begin{aligned}
 H_{\text{after F1}} &= \frac{7}{10} \cdot H_{F1=1} + \frac{3}{10} \cdot H_{F1=0} \\
 &= 0.7 \times 0.863 + 0.3 \times 0.918 \\
 &= 0.604 + 0.275 = 0.879
 \end{aligned}$$

Information Gain for F1

$$\begin{aligned}
 \text{Info Gain}_{F1} &= H_{\text{parent}} - H_{\text{after F1}} \\
 &= 0.971 - 0.879 = 0.092
 \end{aligned}$$

F2 Split Analysis

Split based on F2:

F2 = 1 (students: 01, 05, 06, 07, 08, 09):

- Grade A: 2 (students 06, 09)
- Grade B: 4 (students 01, 05, 07, 08)
- Total: 6 students
- $p_A = \frac{2}{6} = 0.333$, $p_B = \frac{4}{6} = 0.667$

$$\begin{aligned}
 H_{F2=1} &= -(0.333) \log_2(0.333) - (0.667) \log_2(0.667) \\
 &= -(0.333)(-1.585) - (0.667)(-0.585) \\
 &= 0.528 + 0.390 = 0.918
 \end{aligned}$$

F2 = 0 (students: 02, 03, 04, 10):

- Grade A: 2 (students 04, 10)
- Grade B: 2 (students 02, 03)
- Total: 4 students
- $p_A = \frac{2}{4} = 0.5$, $p_B = \frac{2}{4} = 0.5$

$$\begin{aligned}
 H_{F2=0} &= -(0.5) \log_2(0.5) - (0.5) \log_2(0.5) \\
 &= -(0.5)(-1.0) - (0.5)(-1.0) \\
 &= 0.5 + 0.5 = 1.0
 \end{aligned}$$

Weighted Average Entropy After F2 Split

$$\begin{aligned}
 H_{\text{after F2}} &= \frac{6}{10} \cdot H_{F2=1} + \frac{4}{10} \cdot H_{F2=0} \\
 &= 0.6 \times 0.918 + 0.4 \times 1.0 \\
 &= 0.551 + 0.4 = 0.951
 \end{aligned}$$

Information Gain for F2

$$\begin{aligned}
 \text{Info Gain}_{F2} &= H_{\text{parent}} - H_{\text{after F2}} \\
 &= 0.971 - 0.951 = 0.020
 \end{aligned}$$

F3 Split Analysis

Split based on F3:

F3 = 1 (students: 03, 04, 07, 09, 10):

- Grade A: 3 (students 04, 09, 10)
- Grade B: 2 (students 03, 07)
- Total: 5 students
- $p_A = \frac{3}{5} = 0.6$, $p_B = \frac{2}{5} = 0.4$

$$\begin{aligned}
 H_{F3=1} &= -(0.6) \log_2(0.6) - (0.4) \log_2(0.4) \\
 &= -(0.6)(-0.737) - (0.4)(-1.322) \\
 &= 0.442 + 0.529 = 0.971
 \end{aligned}$$

F3 = 0 (students: 01, 02, 05, 06, 08):

- Grade A: 1 (student 06)
- Grade B: 4 (students 01, 02, 05, 08)
- Total: 5 students
- $p_A = \frac{1}{5} = 0.2$, $p_B = \frac{4}{5} = 0.8$

$$\begin{aligned}
H_{F3=0} &= -(0.2) \log_2(0.2) - (0.8) \log_2(0.8) \\
&= -(0.2)(-2.322) - (0.8)(-0.322) \\
&= 0.464 + 0.258 = 0.722
\end{aligned}$$

Weighted Average Entropy After F3 Split

$$\begin{aligned}
H_{\text{after F3}} &= \frac{5}{10} \cdot H_{F3=1} + \frac{5}{10} \cdot H_{F3=0} \\
&= 0.5 \times 0.971 + 0.5 \times 0.722 \\
&= 0.486 + 0.361 = 0.847
\end{aligned}$$

Information Gain for F3

$$\begin{aligned}
\text{Info Gain}_{F3} &= H_{\text{parent}} - H_{\text{after F3}} \\
&= 0.971 - 0.847 = 0.124
\end{aligned}$$

F4 Split Analysis

Split based on F4:

F4 = 1 (students: 02, 04, 05, 06, 07):

- Grade A: 2 (students 04, 06)
- Grade B: 3 (students 02, 05, 07)
- Total: 5 students
- $p_A = \frac{2}{5} = 0.4$, $p_B = \frac{3}{5} = 0.6$

$$\begin{aligned}
H_{F4=1} &= -(0.4) \log_2(0.4) - (0.6) \log_2(0.6) \\
&= -(0.4)(-1.322) - (0.6)(-0.737) \\
&= 0.529 + 0.442 = 0.971
\end{aligned}$$

F4 = 0 (students: 01, 03, 08, 09, 10):

- Grade A: 2 (students 09, 10)
- Grade B: 3 (students 01, 03, 08)
- Total: 5 students
- $p_A = \frac{2}{5} = 0.4$, $p_B = \frac{3}{5} = 0.6$

$$\begin{aligned}
H_{F4=0} &= -(0.4) \log_2(0.4) - (0.6) \log_2(0.6) \\
&= -(0.4)(-1.322) - (0.6)(-0.737) \\
&= 0.529 + 0.442 = 0.971
\end{aligned}$$

Weighted Average Entropy After F4 Split

$$\begin{aligned}
H_{\text{after F4}} &= \frac{5}{10} \cdot H_{F4=1} + \frac{5}{10} \cdot H_{F4=0} \\
&= 0.5 \times 0.971 + 0.5 \times 0.971 \\
&= 0.486 + 0.486 = 0.971
\end{aligned}$$

Information Gain for F4

$$\begin{aligned}\text{Info Gain}_{F4} &= H_{\text{parent}} - H_{\text{after F4}} \\ &= 0.971 - 0.971 = 0.000\end{aligned}$$

F5 Split Analysis

Split based on F5:

F5 = 1 (students: 01, 02, 06, 07, 08, 09):

- Grade A: 2 (students 06, 09)
- Grade B: 4 (students 01, 02, 07, 08)
- Total: 6 students
- $p_A = \frac{2}{6} = 0.333$, $p_B = \frac{4}{6} = 0.667$

$$\begin{aligned}H_{F5=1} &= -(0.333) \log_2(0.333) - (0.667) \log_2(0.667) \\ &= -(0.333)(-1.585) - (0.667)(-0.585) \\ &= 0.528 + 0.390 = 0.918\end{aligned}$$

F5 = 0 (students: 03, 04, 05, 10):

- Grade A: 2 (students 04, 10)
- Grade B: 2 (students 03, 05)
- Total: 4 students
- $p_A = \frac{2}{4} = 0.5$, $p_B = \frac{2}{4} = 0.5$

$$\begin{aligned}H_{F5=0} &= -(0.5) \log_2(0.5) - (0.5) \log_2(0.5) \\ &= -(0.5)(-1.0) - (0.5)(-1.0) \\ &= 0.5 + 0.5 = 1.0\end{aligned}$$

Weighted Average Entropy After F5 Split

$$\begin{aligned}H_{\text{after F5}} &= \frac{6}{10} \cdot H_{F5=1} + \frac{4}{10} \cdot H_{F5=0} \\ &= 0.6 \times 0.918 + 0.4 \times 1.0 \\ &= 0.551 + 0.4 = 0.951\end{aligned}$$

Information Gain for F5

$$\begin{aligned}\text{Info Gain}_{F5} &= H_{\text{parent}} - H_{\text{after F5}} \\ &= 0.971 - 0.951 = 0.020\end{aligned}$$

Summary and Conclusion

Information gain for each feature:

- F1: 0.092
- F2: 0.020
- F3: **0.124**
- F4: 0.000
- F5: 0.020

Conclusion: Feature F3 has the highest information gain (0.124), making it the best feature to split the root node. F3 provides the most information about the grade classification, reducing the entropy from 0.971 to 0.847. Feature F4 provides no information gain at all, as both child nodes have the same entropy as the parent node.

Submission Instruction

Please submit four files to Canvas:

- (i) Submit `hw3.pdf` — it should contain your answers to all questions in this document. A LaTeX template `hw3Latex.txt` is provided.
- (ii) Submit `'hw3lr.py'` — it should be the code that generates both curves in Figure 1.
- (iii) Submit `'hw3knn.py'` — it should be the code that generates Figure 2.
- (iv) Submit `'hw3cknn.py'` — it should be the code that generates Figure 3.